



Antiphospholipid syndrome

Objectives

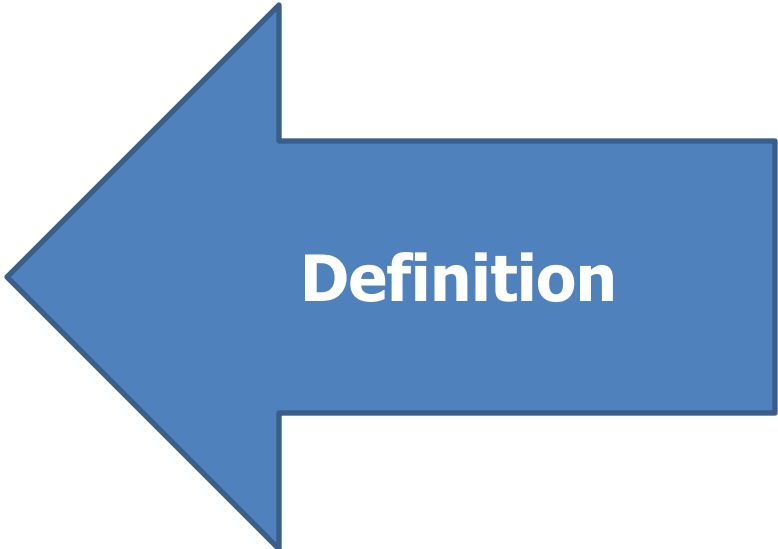
By the end of this lecture students should be able to :

- Define what Antiphospholipid Syndrome is.
- Describe the pathophysiology of APS.
- Identify the diagnostic criteria of APS.
- Outline the management principles for APS.



Is an autoimmune thrombophilic disorder characterized by the presence of antiphospholipid antibodies that increases the risk of thrombosis and adverse pregnancy outcomes.

It may occur as a primary condition or secondary to another autoimmune disease , most commonly systemic lupus erythematosus (SLE)

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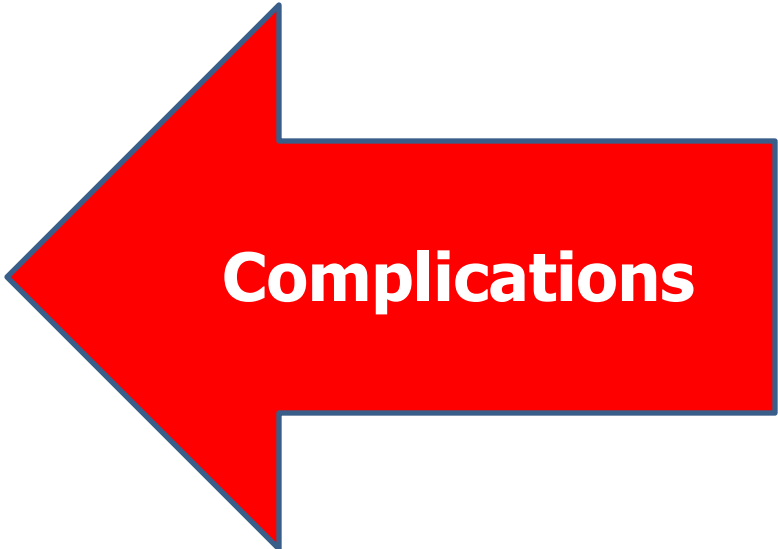
Definition

Pathophysiology

The condition involves autoimmune antibodies directed against phospholipid-binding proteins such as B2-glycoprotein I and cardiolipin. These antibodies promote endothelial injury, platelet activation and complement activation, creating a hypercoagulable state. During pregnancy, this disrupts trophoblastic function and placental circulation, resulting in recurrent pregnancy loss and other complications.

Obstetric complications:

- Recurrent miscarriage (especially beyond 10 weeks)
- Unexplained intrauterine fetal demise(IUFD)
- Pre-eclampsia and eclampsia (often early or severe)
- Fetal growth restriction(placental insufficiency)
- Preterm birth due to severe pre-eclampsia or placental disease
- Placental abruption
- Stillbirth or neonatal morbidity linked to placental thrombosis

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Complications

Diagnosis

Requires at least one clinical criterion and one laboratory criterion , confirmed on two occasions at least 12 weeks apart.

Clinical criteria:

- 1.Vascular thrombosis- one or more episodes of arterial , venous ,or small vessel thrombosis
- 2.Pregnancy morbidity includes:
 - One or more unexplained deaths of a morphologically normal fetus at or beyond 10 weeks , or
 - One or more premature births (<34 weeks) due to eclampsia , severe pre-eclampsia or placental insufficiency ,or
 - Three or more consecutive unexplained miscarriages before 10 weeks

Laboratory criteria:

- Lupus anticoagulant positivity
- Anticardiolipin antibody(aCL) of IgG or IgM isotype at moderate/high titre
- Anti-B2 glycoprotein I antibody of IgG or IgM isotype.

Management of future pregnancy

Preconception care:

- Counsel regarding risk of miscarriage , stillbirth, and pre-eclampsia
- Optimize control of any co-existing autoimmune disease (e.g SLE)
- Discontinue teratogenic medications (e.g warfarin): switch to LMWH preconception
- Screen and address modifiable risk factors : smoking , obesity , hypertension

Antenatal management:

- Start low dose aspirin (75-150 mg daily)preconception or by 12 weeks.
- Add prophylactic or therapeutic LMWH depending on history of thrombosis.
- Close monitoring with growth scans, uterine artery Doppler , and BP/proteinuria checks.
- Monitor for pre-eclampsia , fetal growth restriction , and placental insufficiency

Delivery planning:

- Aim for term delivery if maternal and fetal condition stable.
- Stop LMWH 24 hours prior to induction or caesarean section
- Multidisciplinary care involving obstetrician , hematologist , and anesthetist.

Postpartum management

- Continue LMWH for at least 6 weeks postpartum or transition to warfarin if indicated.
- Provide thromboprophylaxis and monitor for postpartum thrombosis
- Offer counseling for future pregnancy planning and contraception .

Conclusion

APS is an autoimmune prothrombotic disorder with timely diagnosis, individualized therapy, and close monitoring, most women with APS can achieve successful pregnancies and good long-term outcomes.

References

Green-top Guideline No. 17: “Recurrent Miscarriage” (June 2023 version) for easy reference.